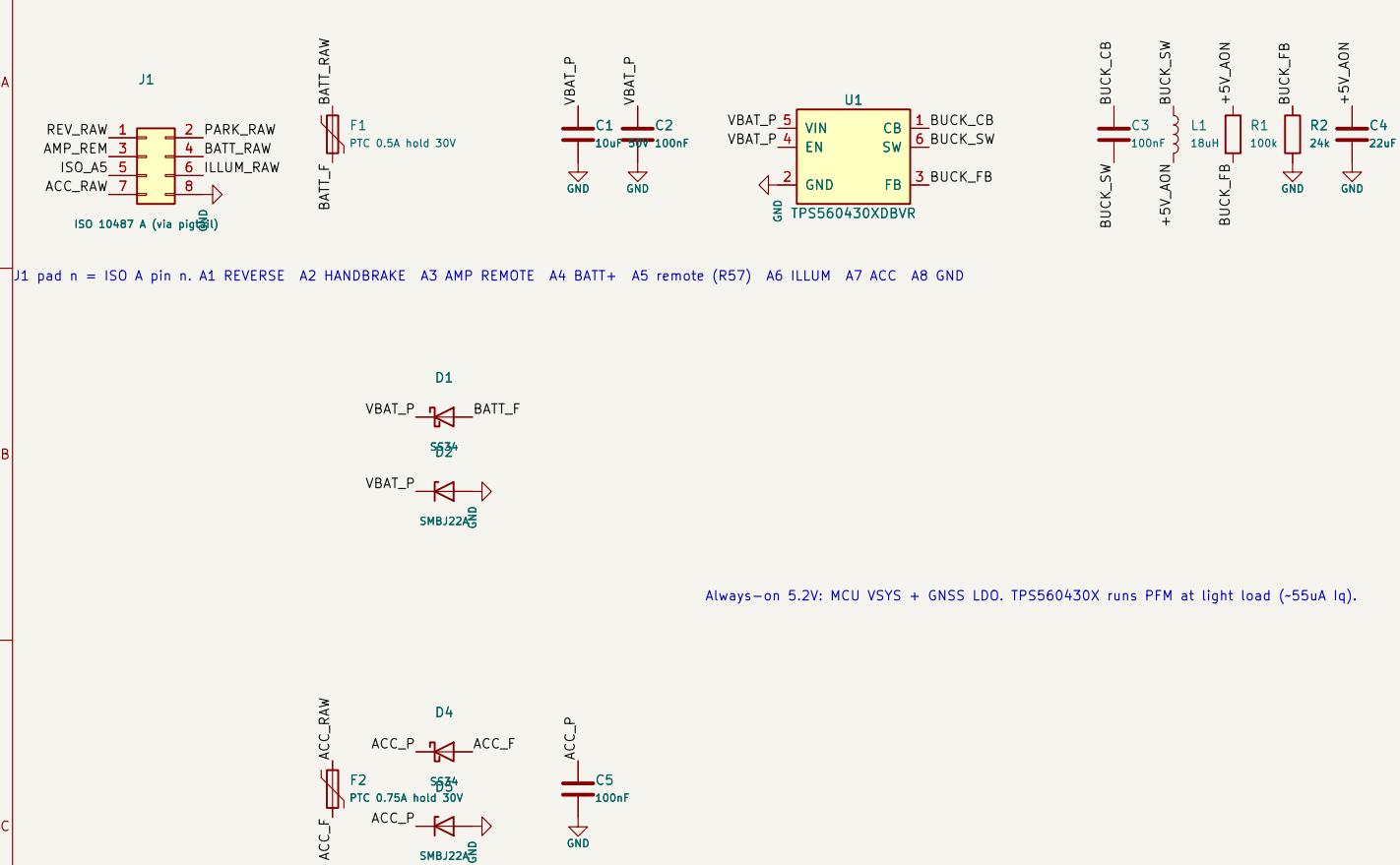
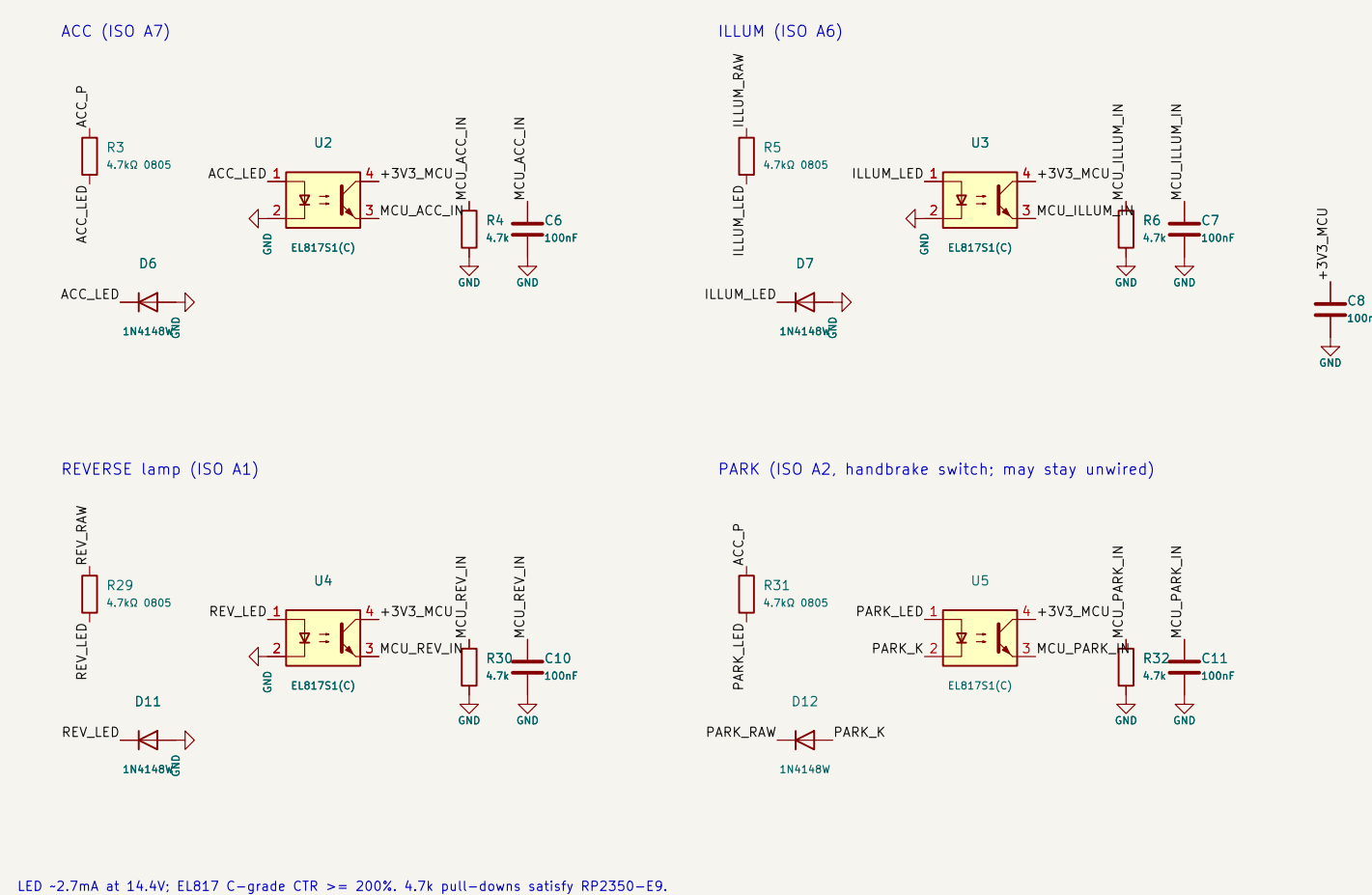


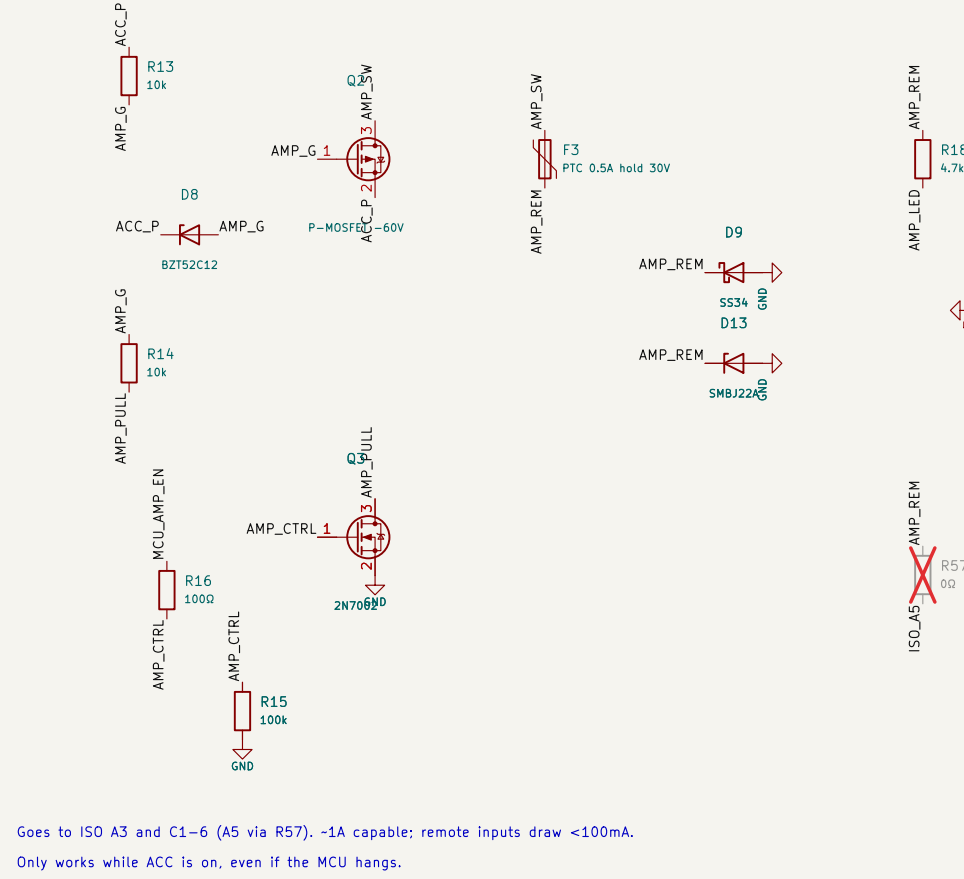
1. ISO 10487 CONNECTOR A (power) + ALWAYS-ON 5V FOR PICO AND GNSS



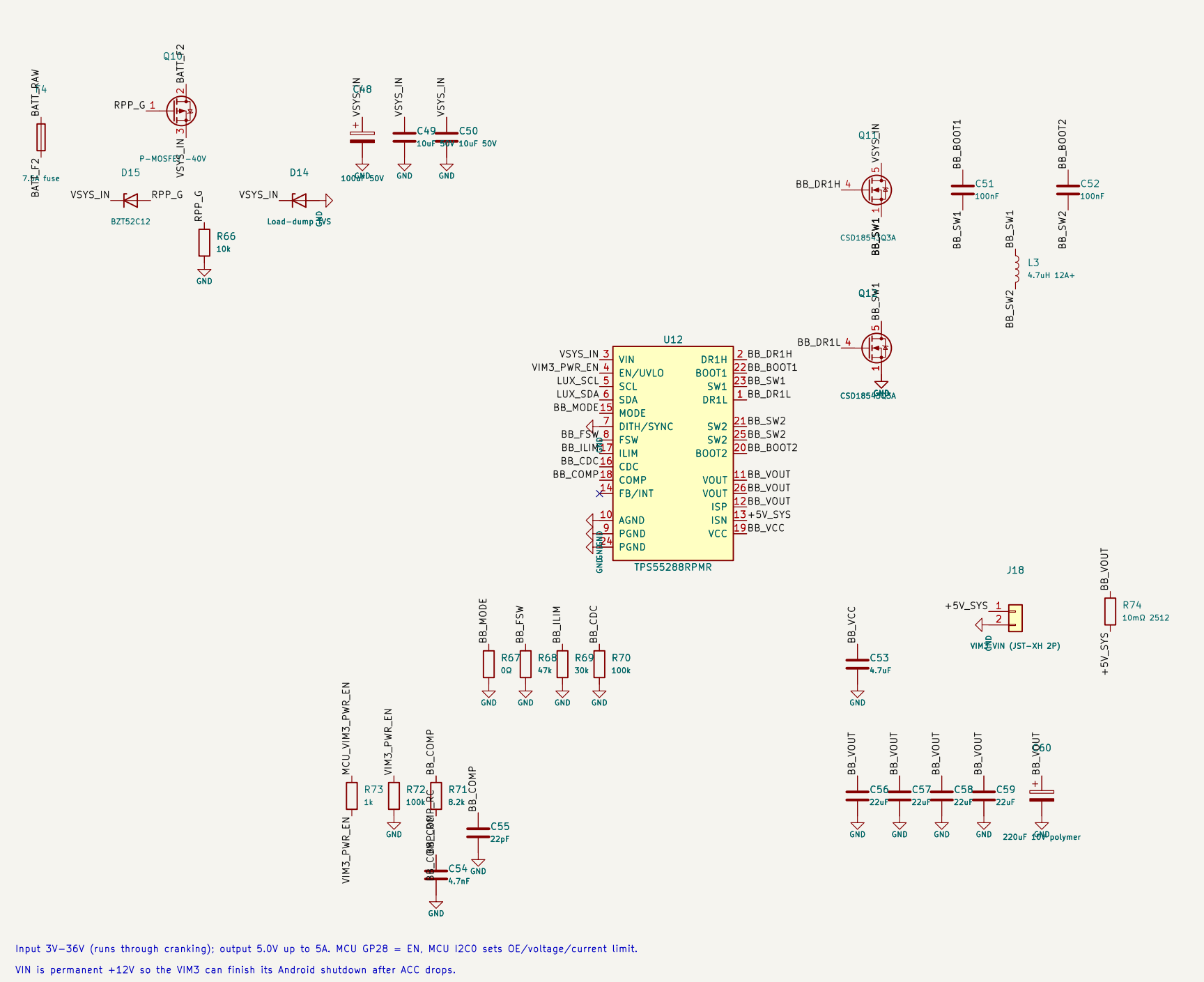
2. VEHICLE INPUTS (opto-isolated). ACC, ILLUM, REVERSE: +12V = on. PARK: switch to ground = on.



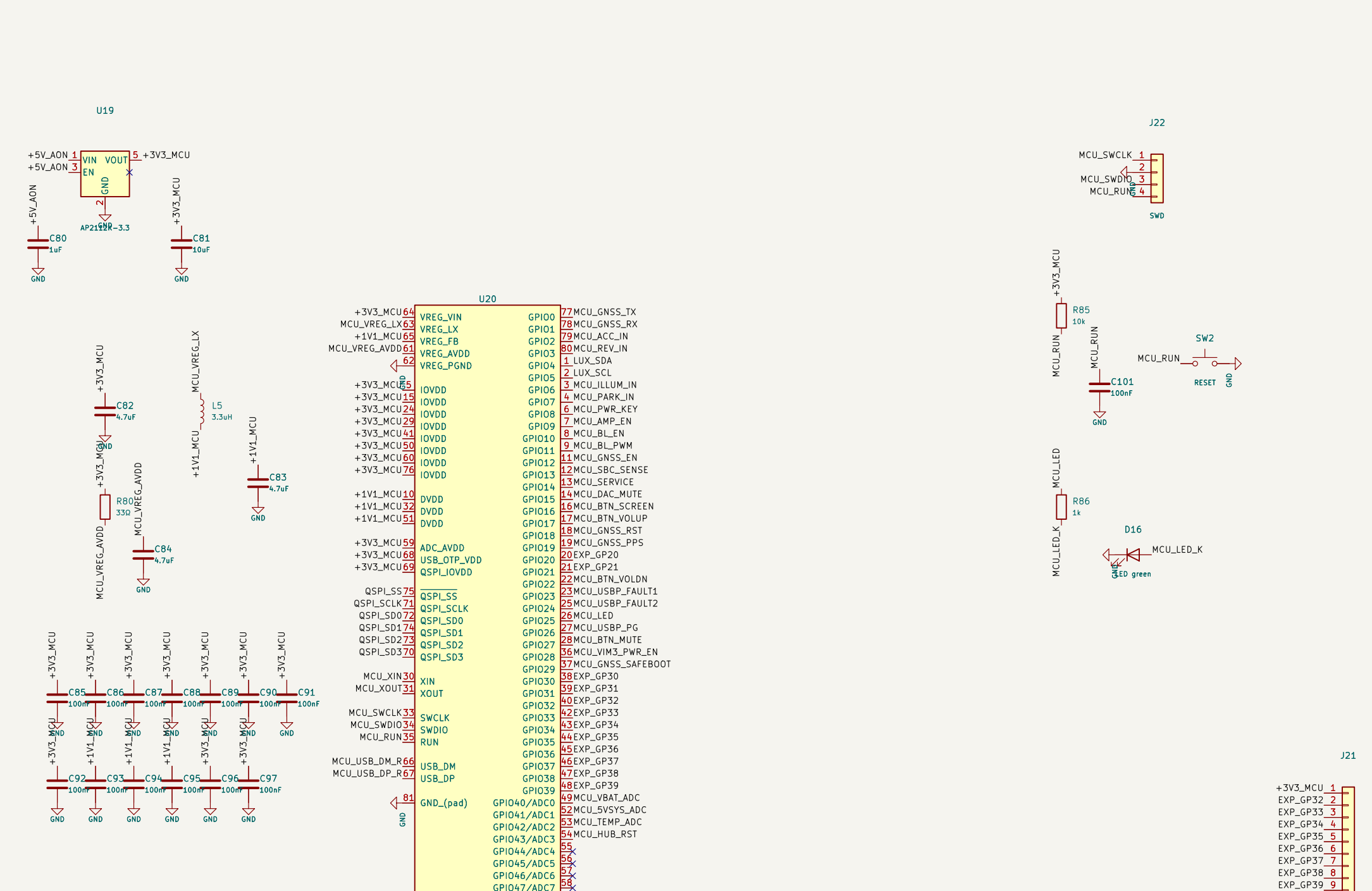
3. AMPLIFIER REMOTE (+12V from ACC, MCU GP9)



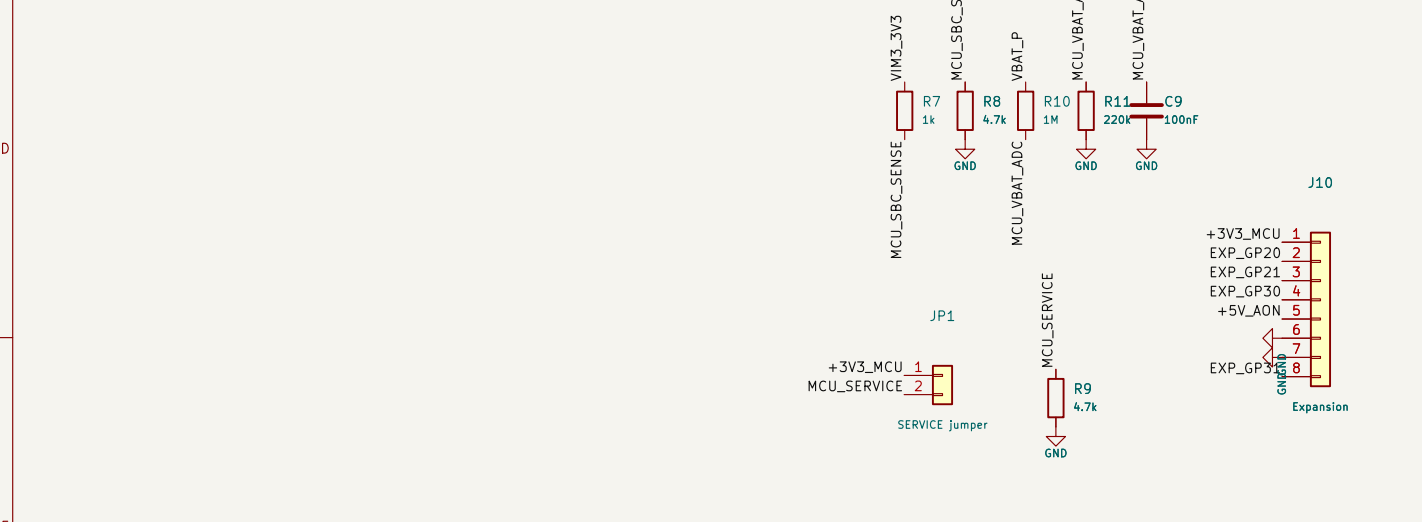
10. 5V/5A SYSTEM SUPPLY FOR VIN3 + SCREEN + HUB (TPS55288 buck-boost, rides through cranking and overvoltage)



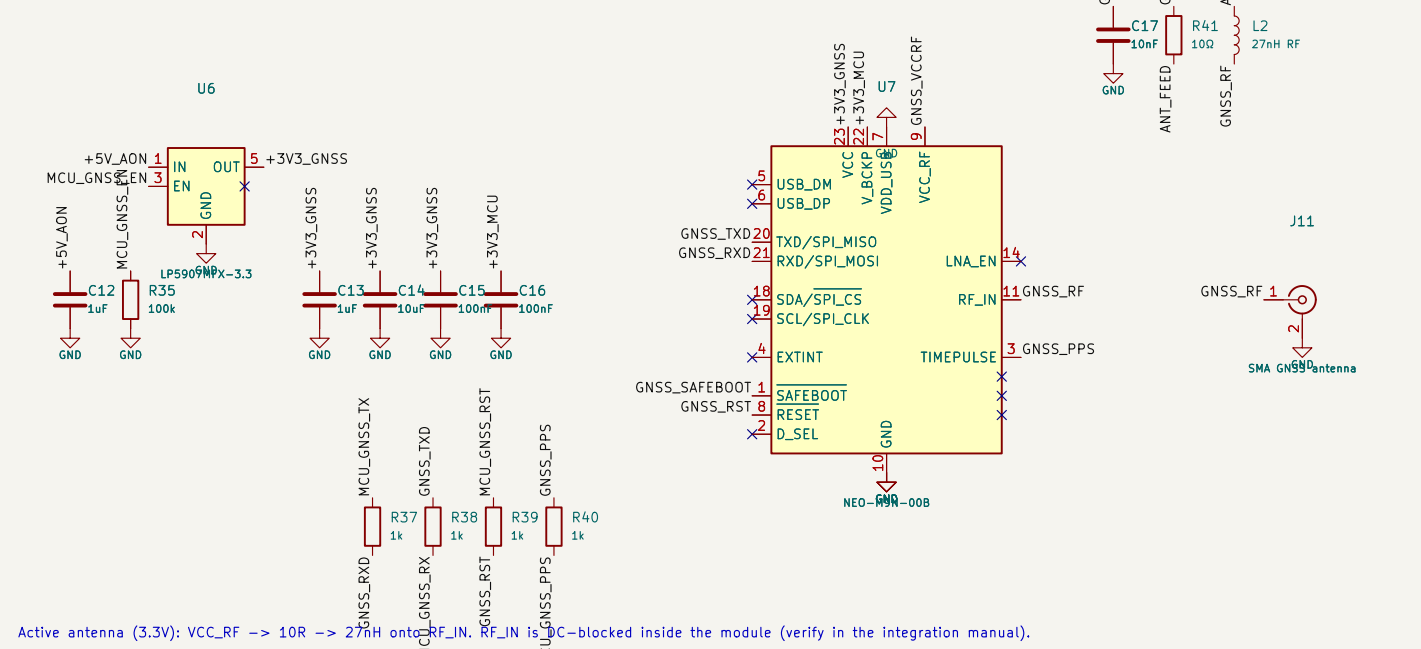
13. MCU: RP2350B (QFN-80) + 16MB QSPI flash + 12MHz crystal + 3.3V LDO, per the Raspberry Pi Pico 2 reference design



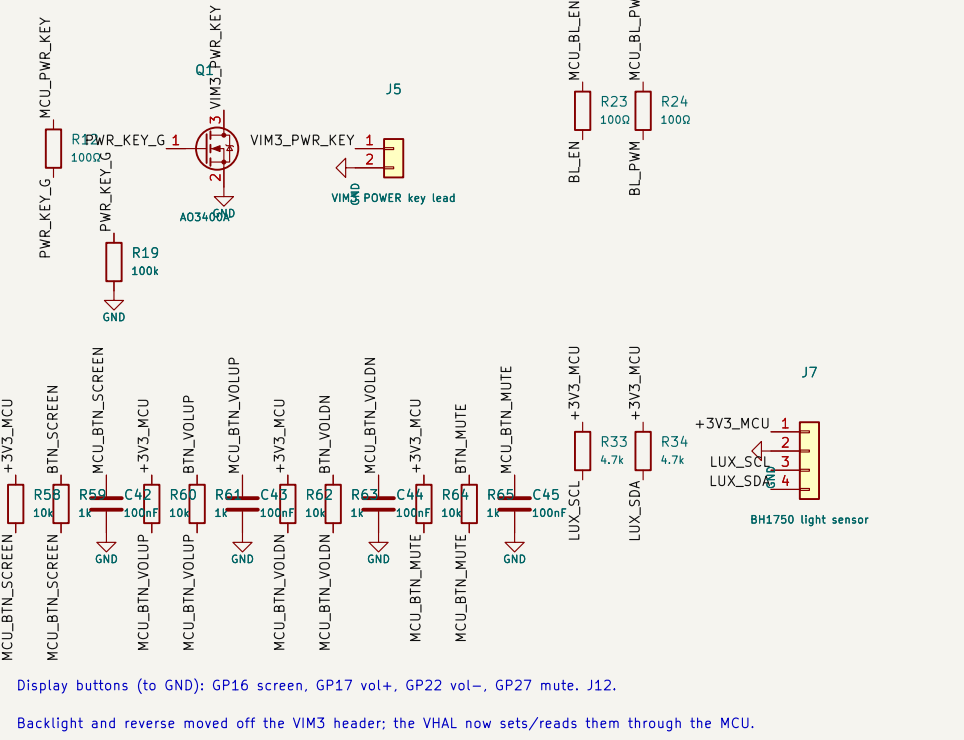
4. MCU HELPERS (MCU core is section 13)



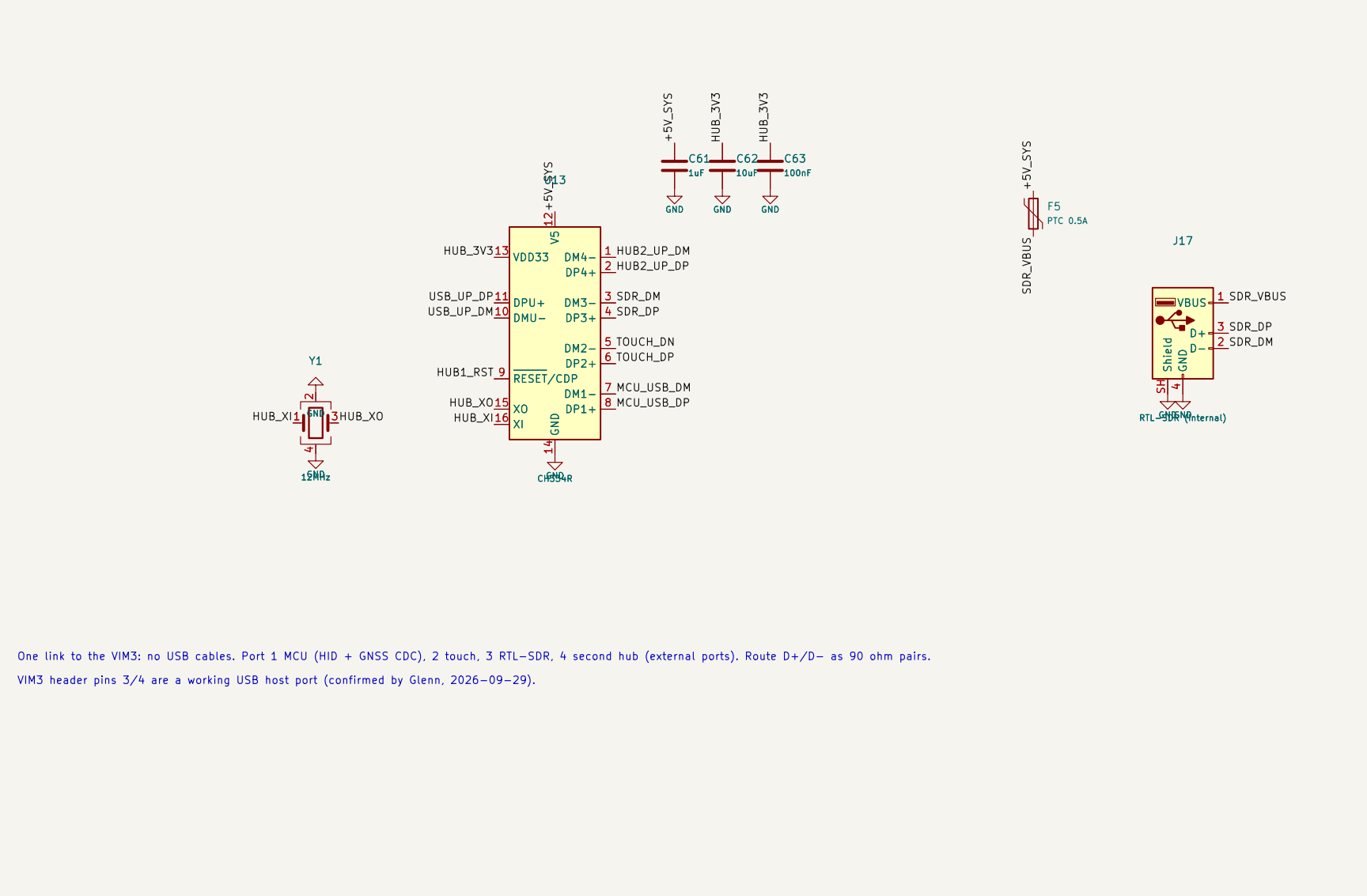
5. GNSS: u-blox NEO-M9N, active antenna on SMA



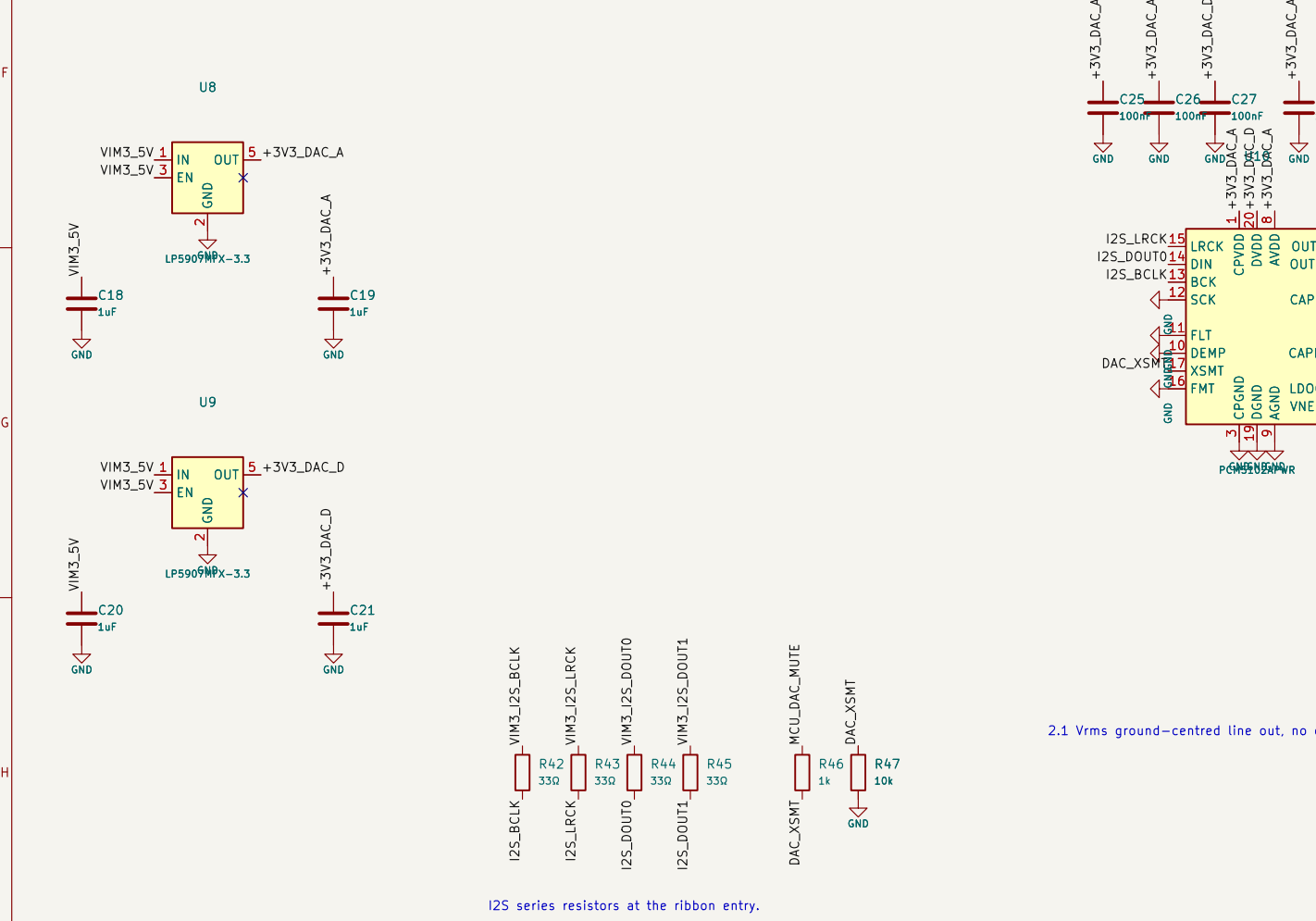
6. VIM3 POWER KEY, BACKLIGHT, LIGHT SENSOR (all from the MCU)



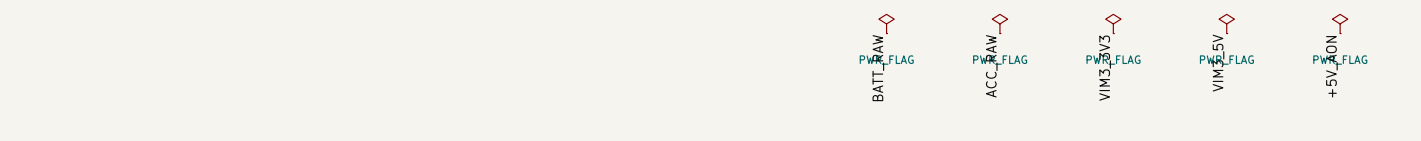
11. USB HUB: VIM3 header USB (pins 3/4) -> MCU, screen touch, spare port



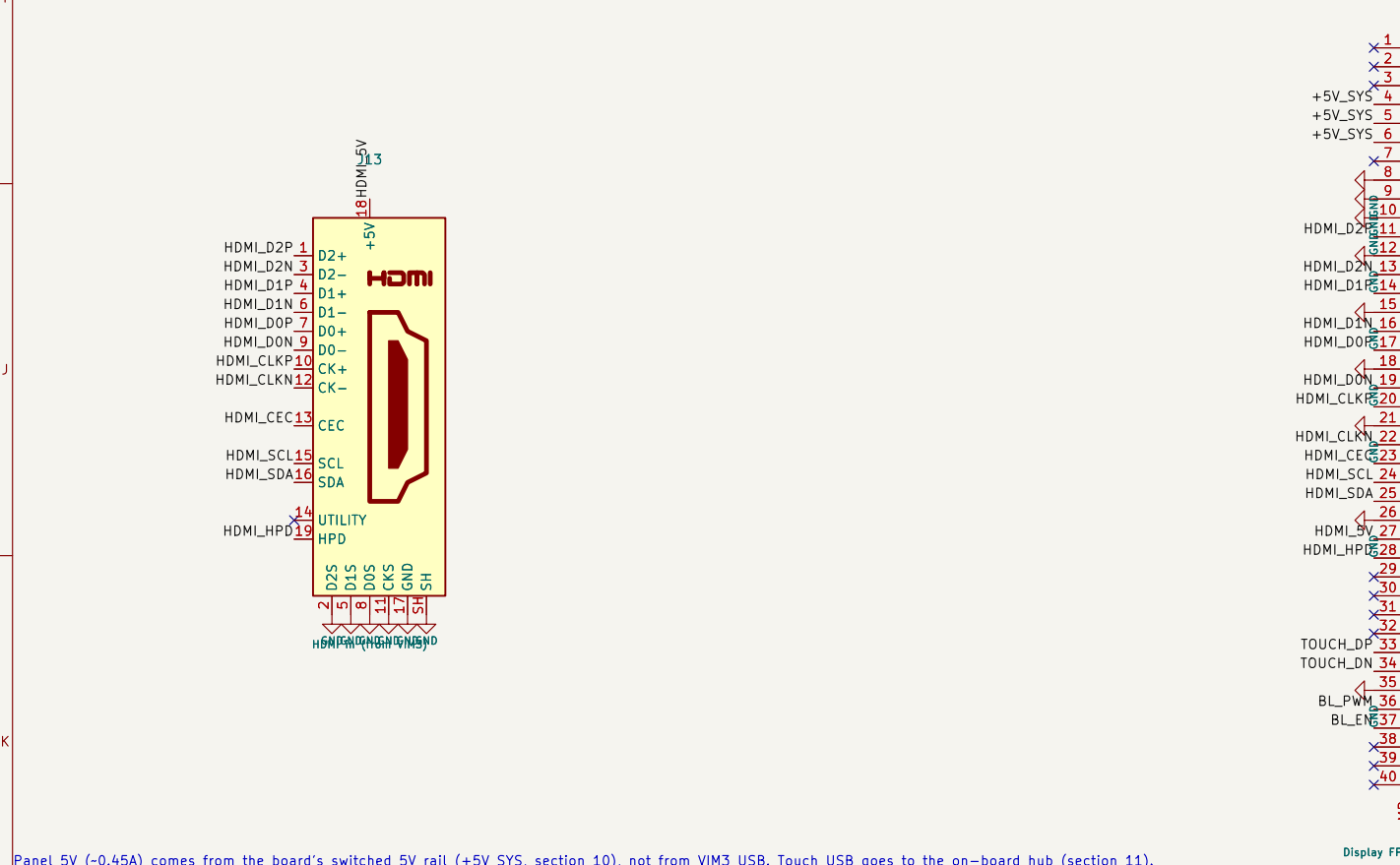
7. AUDIO: VM3 TDM-B I2S -> 2x PCM5102A (front + rear) -> line out on ISO C1



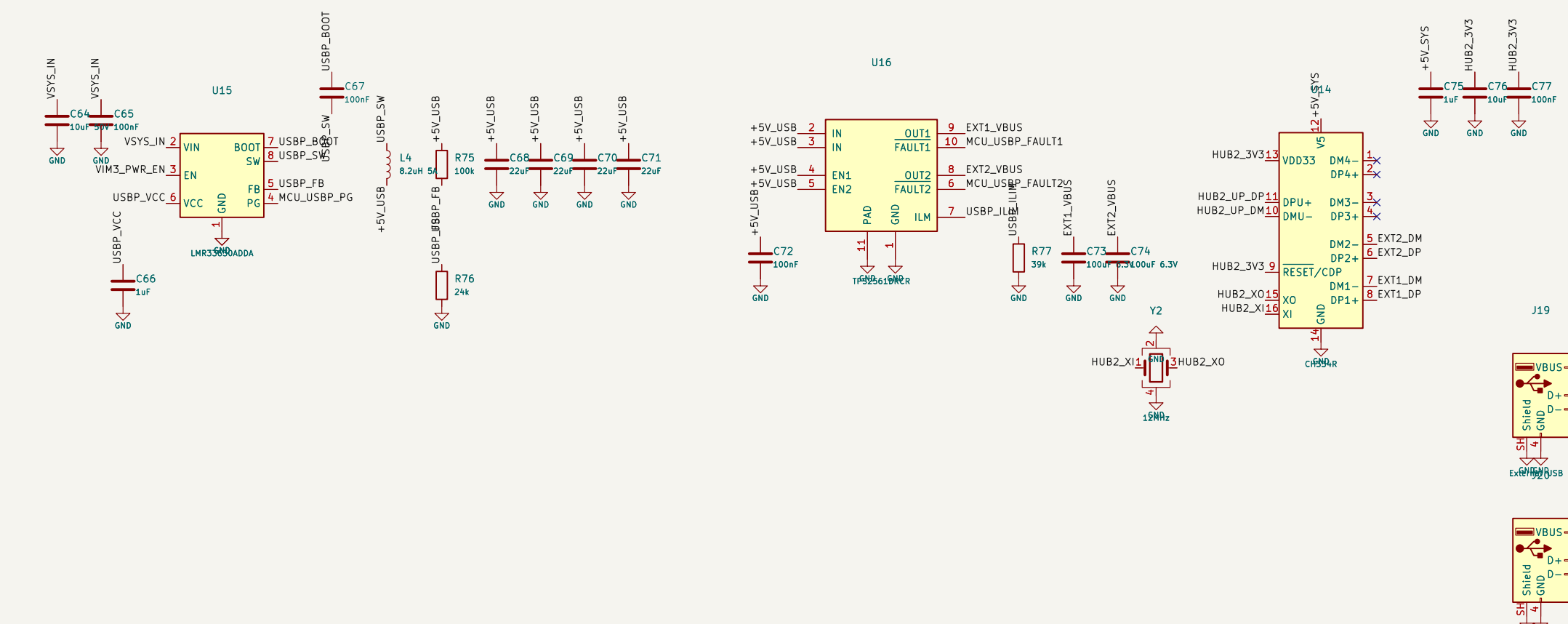
24-bit to 32-bit I2S clock, 48/96/192 kHz.



9. DISPLAY: Waveshare 7in 70H-1024600 adapter built in (HDMI in from VIN3 -> 40-pin FPC to the panel)



12. EXTERNAL USB PORTS (phone etc.): second hub + dedicated 5V/3A supply, 1.5A current-limited per port



Panel 5V (<0.65A) comes from the board's switched 5V rail, (+5V_SYS, section 10), not from VM3 USB. Touch USB goes to the on-board hub (section 11).

Panel SW (+0.45A) comes from the board's switched 5V rail (+5V_SYS, section 10), not from VM3 USB. Touch USB goes to the on-board hub (section 11). Backlight EN/PWM now go straight to the panel: 0V = brightest, 3.3V = darkest (invert PWM in firmware). J6 removed.

IPC orientation as on the WaveShare adapter: seen from the top, ribbon leaving the board edge, pin 1 is on the RIGHT, pin 40 on the left.